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The Planet Is a Wrap, the Planetary Is Computation and Its Gaps

Jussi Parikka[1,2][a]

[1] Digital Design and Information Studies, Aarhus University, Aarhus, Denmark, [2] Winchester School of Art, University of Southampton, Southampton, UK

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This essay addresses planetarity through planetary computation, a term with various uses in design and contemporary philosophy of technology. The text unfolds through the example of the *Venta Mærsk* ship that in 2018 traveled the so-called Northeast Passage, carrying among its cargo frozen fish. The theme of the artificial cryosphere represents one example of logistical media of planetarity, while the essay expands to discuss planetary computation as creation of space in the context of climate change, models, and what the text refers to as epistemic, spatial "wraps" that define possible trajectories of movement and transport within a world of radically changing temperatures and disappearing environments such as ice and snow.

The planet is a constellation of corridors, gateways, hubs, spokes, and terminals. In other words, the planet is made of logistical media that constitutes a significant element of technological power as the control of flows of materials and information.[1] Such a point is rather obvious by the 2020s, after the supply chain glitches during a worldwide pandemic and at the back of military operations such as the Russian invasion of Ukraine (and subsequent links to energy or wheat provision) or the Huthi rebel attacks on ships in the Red Sea, or the tariff wars that spark the ups and downs of prospective plans of what can, should, could, or will not be shipped across economic zones.

Is it true that infrastructures remain invisible until they fail (Star 1999)? Probably yes, although some are invisible exactly because they are programmed to be so. Some absences and disappearances are planned.

Rewind to 2018. The container ship *Venta Mærsk* sails from Vladivostok, Russia, on August 22, arriving at another Russian port, in St. Petersburg at the end of the Gulf of Finland, on August 28. Crossing through the usually ice-covered sea of the Northern Sea Route—or the Northeast Passage—the ship carries hundreds of containers of frozen fish and electronics, two emblematic products of twenty-first-century logistical routes at the center of major geopolitical conflicts in their own right (food and technology). Slightly earlier, a liquid natural gas tanker had completed a voyage through the same passage, and so had the cargo ship *Hyundai Glovis*. However, the Danish shipping giant's vessel is widely advertised in media and the company's marketing as the first container ship to make use of the channel without icebreakers.

The trip was mostly a trial-run, future-planning experiment that tested the viability of the route. The exact route of the ship is still not even fully known as the ship's transponder was switched off part of the way, probably as part of the corporate data gathering related to the most viable corridor available. It links and contrasts well with nineteenth-century historical cases such as the infamous HMS *Terror* and HMS *Erebus* (the Franklin expedition) voyages and tropes of ice, coldness, preservation, movement, and stalling of movement due to the natural effect of what lowering temperatures do to water (Geiger and Beattie 1987).

The *Venta*, though, pushed through the Northeast Passage with its inside frozen and outside melted due to intensifying patterns of climate change opening up new logistical routes. The case is exemplary of planetary-scale computation and logistical cartography that defines the surface of the planet as prepared for industrial transport. What's more, it highlights the regular gaps in sensing planetary objects (Gabrys 2016) that also push back on the otherwise hegemonic-seeming remote-sensing apparatus that wraps the planet into different spheres of control and command.

Such wraps define the partial perspectives of sensing but also the modeling of the planet and its material spaces. Wraps can be considered as models, but they can also be more concrete, like software environments in which the model takes place. The wrap is epistemic as it sparks pos-

a Correspondence: parikka@cc.au.dk
https://pure.au.dk/portal/en/persons/parikka%40cc.au.dk

1 On logistical media, see Rossiter (2021, 132–50).

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sible knowledge, and knowledge sparks further action. The wrap is a fun concept also for the reason that it is spatial in its own right: it is spatial, and it produces space at multiple scales.[2]

This example speaks to the link not only between computation and logistics, but also between elemental media and climate change. Any ship that traverses the opening up of the route also amplifies the existence of the ice-free route. At the same time, the monitoring of dissipated ice turns into models of prospective logistical pathways. The Mærsk melting machine[3] performs the computational event, where ice, models and navigational paths coincide.

In short, planetary computation is fundamentally about multiscalar recursive processes that are informational and material. The *Venta* is thus an excellent model object to understand planetary computation. Or in other words: such a model helps to discuss how planetary computation—and materiality of planetarity—is defined by the patches of disappearance and gaps, planned and unplanned.

II

Planetary computation is a term that features in recent discussions about large-scale computational infrastructures and the double role computation, digital culture, plays in this scheme concerning the planet(ary). The overlap with the technosphere discussion is significant. For example, Benjamin Bratton (2022) refers to "a planetary-scale technostructure of computation—an almost inconceivably vast and complex interlocking system (or system of systems) of sensors, satellites, cables, communications protocols and software" that becomes also an epistemic modeling of that earth which it stems from. Bratton continues: "The development of this structure reveals and deepens our fundamental condition of planetarity—the techno-mediated self-awareness of the inescapability of our embeddedness in an Earth-spanning biogeochemical system that is undergoing severe disruptions from the relative stability of the previous ten millennia" (Bratton 2022). Such a stance feeds into the Antikythera research program, too, that Bratton has set up as a think tank type of platform for design, philosophy, and computation.[4]

I have referred to the double role of media and materiality as "medianatures," a term that picks up on Donna Haraway's and others' use of "natureculture" (Parikka 2015). Medianatures is the recursive zigzag between techniques of sensing and modeling and the materials that make this sensing, modeling, and computation possible. They are directly about the minerals, energy, and other materials that are conditions of computation—hence the punch line that computing is linked to earth's planetary history. Medianatures also extends, though, to such elemental media conditions as the models constructed in technological modes of knowing. This is not to be mistaken as a version of the idealization of computational sensing that produces worlds without friction but the friction of materiality as the condition of computing and sensing (Rossiter 2021, 145). Planetary computation and sensing includes the disappearance of ships and sea ice, of conditions of habitability.

Besides materials of production, medianatures can also address the material environments in which computational procedures hit the ground, in a manner well addressed by Jennifer Gabrys: one can speak of such technogeographies of the computational planet in the active sense (Gabrys 2016). The computational planet is a planet in the middle voice: processes generate subjects and objects, and computing and sensing are no different as they generate a sense of the conditions that they are also reliant upon. It is in the midst of such technological assemblages that the conditions of those assemblages are formulated as objects that are known and actionable. This could easily be seen as a version of the critical theory school's focus on "instrumentalization" or, even worse, Heideggerian "standing-in-reserve," but there is something more interesting at stake. It is, after all, not just the extractivist modes of instrumentalizing resources that are at stake, but also, potentially, forms of sustainable planning. In short, societies also need this computational infrastructure and its capacities to understand multiscalar material interactions.

This does not lessen the political stakes. The political economy of such planetarity and its advanced technological infrastructures are also infrastructures of logistical targeting, extractivism, and the militarized security apparatus of planetary governance. Yet the large-scale implications cannot also be reduced to such, either.[5] We are left with the question: What models of computation are feeding forward an intellectually or ethically satisfactory model of the planetary? At the moment, we are not close to either of these.

Importantly, we can refer to this scheme as recursive datafication where the material agency of sensing and computation transforms territories in which such sensing and computation will further amplify its own repercussions. Like the Northeast Passage–faring *Venta*, the recursive traits of logistics and computation inscribe their own impact upon concrete elemental landscapes—soil, water, ice,

2 My discussion of the wrap is in debt to McCormack (2018). McCormack draws on Michel Serres's use of the "wrap" of representation around things but teases out a more enabling kind of an understanding for such wraps as cultural techniques that do not merely constrain but also produce. "And yet, across so many domains of life it is the very process of wrapping and enveloping that generates a relation of difference through a process of partial enclosure" (113).

3 A specific thanks to Abelardo Gil-Fournier for this formulation.

4 Antikythera think tank, https://antikythera.org/, and journal, https://journal.antikythera.org/.

5 In Bratton's words, "technology transforms the horizon not just of action but of what forms of reality can and cannot be mediated and comprehended," including the complexity of modeling itself as part of the drill. Bratton (2025, 21).

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atmosphere—while also making sense of those environments. In Bernhard Siegert's words, speaking of cultural techniques of making modern maritime space, the point concerns that which occupies space as a way of making it: "Space in general does not exist independently of cultural techniques of space control, and maritime space in particular cannot be separated from the practice of navigation that has disclosed maritime space in the first place" (Siegert 2007, 7).

Siegert continues: "The chorein of the ships is an act of setting up space, which precedes place and founds the capability-to-be-at-a-place; it is the violence of an original space seizure in placeless realms, which is so terrible because of its immemoriability, a vectorial affect that causes a first striation of the smooth space. The terrible chorein, which introduces a spatiality and a being of things outside the polis, which from the perspective of the polis has to appear as eerie, ghastly, and without rights, constitutes the possibility to rule the placeless as maritime space, to found colonies, to visit foreign coasts, to degustate and trade this year's wine, or to subdue other peoples" (Siegert 2007, 7).

Here, a *chorein of planetary computation* is not merely the physical ship and its material existence but the entire logistical apparatus of modeling and sensing that precedes place but also precedes voyages, routes, corridors, ice, atmospheres as they are mapped and projected in perpetuity. In other words, chorein produces the spatiality of the planet in multiple time frames of pasts and futures. But in our case of the *Venta*—or the shadow fleet of Russian ships, including the detection-avoiding military ships regularly reported in the news in 2025—the chorein is also a setting up of space in ways that escape detection and being locatable. It is not just the ship but the environment that is remodeled, spoofed, camouflaged, reinvented in such techniques and infrastructures of planetary computation.

This computational realism is premised on disappearance as its aesthetic register—not seeing but asking what is missing, what needed to be excluded or what was omitted. Earlier, this was the central element of film—as much defined by the gaps between frames as by the frames (Cubitt 2023, 1–24; Virilio 1991). Now, it is the signal space of sensing. It is also the failing to sense, models that project, or models trying to catch up with the real. If the *Flying Dutchman* marked a particular ghostly occupation of the East India Company period of producing the illusion of smooth global space, then the *Venta* and other disappearing ships in their signal space of planetary computation are the twenty-first-century equivalents.

Did the ship disappear, though, or did the environment around it?

III

Beyond the cityscape, the expanded planetary urbanism becomes a space of creating space (Brenner 2013). This geographical scope, though, includes not only territories of dry land but also water and oceans, space and its celestial objects (Bishop 2025), and other realms that also push the computational to new realms and capacities of processing (simulating, modeling, analyzing, projecting, and forecasting).

The moving cryosphere of fish traveling across melting landscapes is a powerful image that replicates well some of the critical insights into shipping and logistics. I like to think of it as an unwritten, unfilmed sequel to Allan Sekula and Noël Burch's film *The Forgotten Space* (2010), which focuses on logistics and capitalist maritime space. In their commentary to their film, they emphasize this point about creation of logistical spaces as part of reorganizing planetary surfaces: "Our response to these myths is that the sea is the key to understanding globalized industrialism. Without a thoroughly modern and sophisticated 'revolution' in ocean-going cargo-handling technology, the global factory would not exist, and globalization would not be a burning issue."[6]

While their language speaks of globalization, this can be read in the other register of planetarity, too, as it becomes a narrative about the forgotten and the absent conditions: an emphasis on material contexts of where logistics hits the ground (Neilson and Mezzadra 2019) or watery sea routes, which also pushes back against "uniform epistemic representations" (Gabrys 2018). In Gabrys's take on planetarity as a praxis, drawing also on Gayatri Spivak: "The planetary is not a ground or grounding. Instead, it signals toward an 'inexhaustible diversity of epistemes.' [...] It does not definitively come into being as globes or picture-postcards of floating blue marbles. The planetary is the difference, distance, and duration [...]" (Gabrys 2018).

This difference and distance is, though, wrapped in the double sense as part of planetary computation. Firstly, it is the actual diversity of epistemes as well as diversity of complex material environments that is the site of computation as actually existing material infrastructures and operations. For example, this is the sea ice, and it is the disappearance of ice, and the computational sensing and models that try to map this disappearance. Secondly, difference and duration are part of the computational; the material epistemologies of planetary conditions are wrapped inside computerized models and simulations. In another context, I have referred to this as wrapping of elemental media in computational worlds of design and logistics—for example, wind simulation and digital twinning (Parikka, forthcoming; see also McCormack 2018, 29). This wrap is somewhat

6 Sekula and Burch (2010). Christina Sharpe has, however, noted an important critical point in relation to Sekula and Burch. The treatment of the only Black woman in the film unveils a implicit collusion with other forgotten containerizations still in operation, such as migrant ships, which returns the material trope of earlier vessels for enslaved people but also now in the context of global warming and climate change. Sharpe (2016, 25–67).

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like the *Umwelt* in Ryan Bishop's (2025) take in his article in this special collection but also very concretely identified in the software-based computational design and modeling that are partial folds of planetary materials.

In short, working through conflict, absence, and material duration should be seen as very much located within the technosphere of computational praxis, wraps, and infrastructures. The stakes of politics of software of planetarity are also about politics of models of planetarity that ask: according to which model is the planet being built and rebuilt as corridors, territories, watery ways, ships that do not show up as public data points even as they model new pathways for later exploitation? Ships temporarily absent from signal space are also part of the remapping of planetary space for future use, resource extraction, and strategic control, and for dissipation, warming, and other visible or invisible changes. The absence from view marks a future absence to come.[7]

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COMPETING INTERESTS

None identified.

AUTHOR BIOGRAPHY

Finnish cultural historian Jussi Parikka is professor in Digital Aesthetics and Culture at Aarhus University, where he is the codirector of the Environmental Media and Aesthetics research program. He also holds an affiliation with University of Southampton (Winchester School of Art). Parikka's recent books include the coauthored *Lab Book* (2022), *Operational Images* (2023), and the coauthored *Living Surfaces: Images, Plants, and Environments of Media* (2024).

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7 Susan Schuppli's video essay *Cold Rights* (2022) addresses the climate justice and ethics questions about rights of ice and right to coldness. Building on indigenous activists' work, Schuppli's work asks whether sea ice has the right not to be broken and melted. See https://susanschuppli.com/COLD-RIGHTS. The video essay was commissioned for the *Weather Engines* exhibition (2022) at Onassis Stegi, Athens.

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